

DUL-92 — Stuck Pipe Historical Knowledge Pack

Synthetic historical offset-well record for RAG, similarity matching, and hazard explanation

IMPORTANT: This document is synthetic demonstration data prepared for the NWIS/SIH prototype. It must not be represented as an authentic confidential OIL record, real incident report, or verified SPE publication.

Well & Event Summary

Field	Value
Well	Duliajan-92 (DUL-92)
Field	Duliajan Field, Upper Assam Shelf
Event depth	2210–2214.5 m MD
Formation	Barail Group Shale
Hazard	Reactive shale / mechanical pipe entrapment
Spud	12-Oct-2022
Target depth	3520 m
Completion	18-Jan-2023
Rig	OIL Rig 14 (2000 HP)

Incident Narrative

During a synthetic drilling scenario, the drill string became mechanically entrapped while pulling out of hole at approximately 2210 m in Barail Shale. Torque increased sharply and overpull was observed. The event is represented as a historical case that NWIS can retrieve when a current well enters a similar formation and shows a similar sensor signature.

Recorded Parameters

Parameter	Historical value	Reference/normal context
Depth	2210–2214.5 m	Event interval
Mud weight	1.16 g/cc	Synthetic event value
Equivalent mud weight	1.24 g/cc	Synthetic reference
Gamma ray	118 API	Reference 45–60 API
Standpipe pressure	2850 psi	Reference ~2100 psi
Torque	28.4 kN·m	Reference ~11.5 kN·m
Smectite	42%	Reference <15%

Mitigation & Outcome

Step	Action
1	Spot 50 bbl Poly-Glycol/high-lubricity pill
2	Soak for 4 hours
3	Use KCl 8% and glycol 4% treatment
4	Pipe freed at approximately 09:15
5	Pump 30 bbl high-viscosity sweep

Step	Action
6	Condition mud to 1.25 g/cc
NPT	36.5 hours
Estimated impact	■45.6 lakh — synthetic demonstration value

NWIS RAG Facts

Search terms: DUL-92, Barail, 2210 m, stuck pipe, reactive shale, smectite, torque 28.4 kN·m, Poly-Glycol, 4-hour soak. A RAG answer should cite this report and clearly distinguish historical evidence from a recommendation for the current well.

Synthetic Time-Series Feature Template

timestamp	depth_m	torque_kNm	rop_m_hr	rpm	wob_kN	spp_psi	mud_weight_gcc	formation
10:00	2205	18.2	14.5	120	18	2400	1.20	Barail Shale
10:05	2206	19.0	14.1	119	18.5	2420	1.20	Barail Shale
10:10	2207	20.5	13.4	118	19	2450	1.20	Barail Shale
10:15	2208	22.8	12.0	116	20	2490	1.21	Barail Shale
10:20	2209	25.4	10.1	114	22	2550	1.21	Barail Shale
10:25	2210	28.0	8.0	112	24	2600	1.22	Barail Shale